

Apoorva Rajan

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SKILLS

- **AI & Data:** Python, SQL, LLMs, LangChain.js, LangGraph.js, OpenAI, Zod
- **Cloud, Platforms & Software Engineering:** Microsoft Fabric, Azure, ArcGIS, REST APIs, Node.js, Express.js, GitHub Actions, CI/CD, Postman
- **Languages, Frameworks & Tools:** TypeScript, JavaScript, React.js, Lit, Lumina, Web Components, Fluent UI, Vite, Redux, Storybook, Vitest, MongoDB, ELK

EXPERIENCE

- **Esri, Environmental Systems Research Institute** Redlands, CA
Software Development Engineer 2 Sep 2024 - Present
 - Building agentic AI workflows for natural-language interaction with geospatial data using LangChain.js, LangGraph.js, and GPT-5.6 model family, including custom agents and schema-linking capabilities.
 - Enabled agents to dynamically select relevant context from shared state based on user intent, allowing custom and out-of-the-box agents to share information across workflows.
 - Integrated a custom AI agent into ArcGIS Maps for Microsoft Fabric, enabling users to add Lakehouse data to maps and perform geospatial analysis through natural-language prompts.
 - Drove the implementation of ArcGIS MapViewer as a Microsoft Fabric workload using TypeScript, Lit-based Lumina, Fluent UI, and Web Components, reaching 90+ active tenants in preview.
 - Designed and developed a data browser to load OneLake datasets into maps and implemented real-time geospatial analytics using Fabric Event Streams, Eventhouse, KQL, and ArcGIS Stream Layers.
- **Microsoft** Cambridge, MA
Software Design Engineer Intern Jan 2024 - Feb 2024
 - Implemented a patient-centric Copilot chatbot web app using React, TypeScript, Fluent components, SCSS, and Figma to help patients navigate clinical trial information.
 - Delivered personalized responses to patient inquiries by integrating Azure Health Bot, Language Studio, GPT-3.5-Turbo, and the ClinicalTrials API.
- **RedBus, Ibibo Group Private Limited** Bangalore, India
Senior Software Engineer, Software Engineer Aug 2019 - Jul 2022
 - Owned the development of mobile web features for the bus ticketing platform collaborating with product managers, engineers and designers and recognized with Trailblazer and Titans of the Quarter awards.
 - Enhanced the Colombia website domain ranking by 192% through core web vitals optimization and improved bus tracking in low-connectivity areas with Progressive Web Application and service workers.
 - Reduced UI development effort by 30% by building reusable components for the in-house component library and maintained system reliability through ELK monitoring and client and server debugging.
 - Drove approximately 80,000 user registrations by coordinating cross-functional development of VaxAlert, a portal notifying users of available COVID-19 vaccine slots at nearby vaccination centers.
- **Publicis Sapient** Bangalore, India
Data Engineer Intern May 2018 - Jul 2018
 - Contributed to the development of a Customer Data Platform using Scala, Apache Spark, SQL, and GCP, transforming raw data into actionable insights on customer behavior and preferences.

EDUCATION

- **University of Massachusetts Amherst** Amherst, MA
Master of Science in Computer Science Sep 2022 - May 2024
 - **Career Developer:** Mentored 100+ students and built data-driven career outcome dashboards.
 - **Research Assistant:** Built a search application for cross-lingual event and argument retrieval
- **Model Engineering College** Ernakulam, India
Bachelor of Technology in Computer Science and Engineering Aug 2015 - May 2019